

Nikhil Rout

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🐙 NikhilRout 🔗 NikhilRout.github.io

Education

Bachelor of Technology, Electronics and Communication Engineering 2022 – 2026
Vellore Institute of Technology, Chennai CGPA: 9.13/10

Research Experience

Research Intern Remote
Vortex GPGPU, ORCAS Lab, UCLA May 2025 – Present

- Developed a Tensor Core Unit (TCU) extension for the open-source RISC-V based Vortex GPGPU, accelerating WMMA operations in deep learning kernels
- Architected and implemented a configurable Microscaling (MX) Fused Dot Product datapath to explore custom block-floating-point formats
- Performed microarchitecture design space explorations, evaluating format-wise shared multipliers, Modulo-4 operand grouping CSA, and threads/warp arithmetic intensity scaling
- Introduced a novel sparse lane mask clock-gating strategy for dynamic power reduction in workloads exhibiting dual-side unstructured sparsity
- Validated TCU FP numerical accuracy by porting the FTTN benchmark to Vortex WMMA API
- Achieved 4-cycle latency at 300 MHz Fmax on the Xilinx U55C FPGA, delivering $\sim 3.1\times$ throughput over the open-source baseline (Berkeley HardFloat) at less than 60% the area cost
- Demonstrated end-to-end speedups on Llama2-inference benchmarks: 9.2x (prefill) and 2.0x (decode)
- Contributed to maintaining GitHub CI/CD, Issues and Pull Requests

Research Publications

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- [1] **Nikhil Rout**, Blaise Tine, “[Ten-Four: An Open-Source Fused Dot Product Unit for Mixed-Precision GPGPU Tensor Cores](#)”. Under Review *International Conference on Computer Design (ICCD)*, 2026.
 - [2] **Nikhil Rout**, Jean Jenifer Nesam J, “[Optimizing RGB to Grayscale, Gaussian Blur and Sobel-Filter operations on FPGAs for reduced dynamic power consumption](#)”. In *Artificial Intelligence For Internet of Things (AIIoT)*, 2024.

Talks and Workshops

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- Investigating Fused Dot Product Microarchitecture for the Vortex Tensor Core Unit Extension
Vortex Workshop and Tutorial at MICRO 2025

Skills and Competencies

Languages: Verilog/SystemVerilog, C/C++, CUDA, Python/PyTorch

Tools: Quartus Prime, Xilinx Vivado/Vitis, Synopsys VCS/DC, TCL, CMake, LLVM, Git, LaTeX

Simulators: Gem5, RISC-V Spike, GPGPU-Sim, Accel-Sim, Vortex SimX, Ramulator

Projects

Number Theoretic Transform Accelerator on FPGA

- Investigated radix-2 DIT and DIF butterfly structures to evaluate optimal dataflow for FFT and NTT hardware implementation
- Designed pipelined NTT/INTT cores with modulo-arithmetic for polynomial multiplication in lattice-based post-quantum cryptography schemes on FPGAs

Honors and Awards

- Student Travel Grant, IEEE/ACM ISCA 2026 (\$1000)
- Student Travel Grant, IEEE/ACM MICRO 2025 (\$580)
- Outstanding Teaching Assistant, VIT Chennai (FPGA System Design 2025)
- Best Paper Award, IEEE AIIoT 2024

References

Dr. Blaise-Pascal Tine, *Assistant Professor, UCLA*

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Dr. Jean Jenifer Nesam J, *Assistant Professor, VIT Chennai*

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